MINISTRY OF ECONOMY
DEPARTMENT OF STATISTICS MALAYSIA

DOSM DATATHON

DOSM Datathon 2026

Blue Lestari: Harnessing Machine Learning and Business Intelligence for Sustainable Marine Tourism Management in Malaysia

by

ZEFA Analytics

Ahmad Afiq Hilmy Bin Haza Hilmy

Fatin Alisha Binti Mohamed Zaini

Zahin Zikri Bin Zawawi

Erza Nabila Binti Mushidy

Under the Guidance of

Assoc Prof Ts Dr Noryanti Muhammad

Centre for Mathematical Sciences

Universiti Malaysia Pahang Al-Sultan Abdullah

Date of Submission

22 September 2026

CONTENTS **PAGE**

Front page 1

Table of contents 2

Introduction 3 - 4

a. Background of the Study

b. Problem Statement

c. Objectives

Literature Review 5 - 7

Methodology 8 - 13

Findings 14 - 22

Output (Dashboard) 22 - 26

Conclusion 27

References 28

INTRODUCTION

Background of the Study

Marine and coastal tourism is one of the main pillar of Malaysia's blue economy. Island destinations along the East Coast of Peninsular Malaysia such as Pulau Redang, Pulau Perhentian, Pulau Tioman, Pulau Tinggi and Sabah, such as Semporna Archipelago, Sipadan, Mataking attract more than 2.5 million domestic and international visitors annually, accounting for a large proportion of national tourism revenues (Tourism Malaysia, 2026).

Hospitality, scuba diving operations, snorkeling tours, boat charters and markets for local handicrafts bring money to these island ecosystems. However, marine tourism is especially dependent on the visual and biological health of its natural capital, in this case healthy reef coral ecosystems.



The economic viability of marine destinations directly reflects underwater environmental quality. When water clarity, marine life abundance, and living coral structures are maintained, visitor satisfaction and willingness-to-pay (WTP) remain high. As global demand for nature-based experiences increases, establishing a sustainable balance between maximizing visitor revenue and preserving fragile marine ecosystems is essential for long-term economic stability.

This background investigation seeks to focus its scope on five key states which include Terengganu, Pahang, Johor, Kedah, and Sabah. These states are characterized by an abundance of marine biological life forms including unique island formations such as those in Malaysia's marine parks in the peninsula as well as those in the megadiverse coral reefs of Malaysian Borneo. They are therefore at the heart of marine tourism in Malaysia.

Problem Statement

Marine tourism has its economic significance. However, there are three operational challenges face the existing destination management of Malaysia.

First things first, Uncontrolled TCC (Tourism Carrying Capacity). The period of holiday time is when the problem of over tourism arises. Uncontrolled boat anchoring, physical interaction of the tourists who are beginners in diving and over-discharge of waste from resorts are overwhelming the environmental carrying capacity of the island environment.

Tourism Management in Reactive Mode Environmental monitoring has always been conducted by retrospective surveys. Once coral damage or collapse of the habitat has been established, the destination has lost its attractiveness leading to permanent damage to tourism livelihoods of the community.

Tourism bodies monitor tourist arrivals and revenue whereas environmental bodies monitor the condition of the reef. Without a platform which links the two for management purposes, policymakers cannot implement effective measures.

Project Objectives

The **Blue Lestari** project resolves these challenges by uniting business intelligence and machine learning into an actionable executive dashboard. The project delivers three core objectives:

1. To quantify marine tourism baselines by using Malaysian tourism and marine data.
2. To develop a Machine Learning coral forecast model used for predicting coral reefs sustainability.
3. To deliver an integrated decision support system by building a BI dashboard that helps multiple authorities connect the dots between marine and tourism.

LITERATURE REVIEW

Marine Tourism and Coral Reef Dependency in Malaysia

Malaysian marine park research has long identified coral reef health as inseparable from tourism value. An early carrying-capacity assessment of Pulau Payar Marine Park in Kedah described a threshold beyond which continued reef use leads to environmental degradation, saturated facilities, and declining visitor satisfaction, and found that tourism development in the park had already leaned toward a mass-tourism model considered inappropriate for a marine park (Lim, 1998). More recent monitoring reinforces this concern: Reef Check Malaysia's *Status of Coral Reefs in Malaysia 2024* report found that national Live Coral Cover fell from 45.9% in 2023 to 44.7% in 2024, with the organization's CEO attributing the decline mainly to unsustainable tourism practices (The Edge Malaysia, 2025). A subsequent Mongabay (2026) report citing the same organization's data found that Malaysia lost approximately one-fifth of its coral reefs within three years, and that even formally protected marine areas showed weak recovery of key fish and invertebrate populations suggesting protection status on paper does not guarantee ecological outcomes. This last point echoes the current study's own EDA finding that "No protection" sites and "Marine Park" sites did not differ as sharply in mean LCC% as protection labelling alone would suggest.

Tourism Carrying Capacity and Reactive Management Gaps

The carrying capacity concept is central to reef tourism management. Ecological carrying capacity has been defined as the ability of a resource to withstand recreational use without incurring unacceptable damage to its ecological components, a concept widely used in diving tourism management despite the absence of any universally accepted quantification method (Zhang et al., 2016). Empirical work on Mabul Island in Sabah found that diver-damaged hard coral increased exponentially with rising diver numbers, and calculated the site's ecological carrying capacity at 15,600 - 16,800 divers per year, concluding that management intervention becomes justified once 8 - 14% of hard coral shows damage (Zhang et al., 2016). This directly supports the present study's problem statement regarding uncontrolled diver contact and boat activity as drivers of reef stress.

Reef Check Malaysia (2025) has separately argued that during peak season, many Malaysian islands already exceed practical infrastructure carrying capacity citing water shortages and electricity blackouts and has called for a shift away from mass tourism toward a joint management model integrating local communities into marine protected area governance. This recommendation reflects the same institutional fragmentation between tourism and environmental authorities that motivates the

Blue Lestari project's integrated BI approach. Willingness-to-pay research at Malaysian marine parks has further shown that visitors place measurable monetary value on reef quality (Ahmad & Hanley, 2009), reinforcing the direct link between ecological condition and tourism revenue underlying this project's core premise.

Machine Learning Applications in Coral Reef Forecasting

Machine learning has increasingly supplemented traditional survey-based reef monitoring because ecological relationships are rarely linear. Arcodia et al. (2025) used an XGBoost model to forecast the onset of moderate coral heat stress on Florida's Coral Reef up to six weeks in advance, with a mean absolute error of approximately one week, outperforming logistic regression and frequency-based baseline models at the reef sites studied. At a global scale, a machine-learning ensemble trained on over 45,000 coral field observations and 42 climate and human-pressure predictors was used to estimate coral cover for 2020 and 2050 across a high-resolution global reef habitat dataset, identifying more than 166,000 km² of climate-resilient reef area across dozens of countries (cited in ResearchGate, 2026). Work applying remote-sensing classifiers in the Red Sea similarly used a Support Vector Machine trained on depth-invariant indices to detect coral cover change over an 18-year period, finding declines ranging from roughly 3% to 14% depending on location (PMC, 2024). These studies establish LCC/reef-condition forecasting as a well-precedented ML application, though most rely on remote sensing or large multi-decadal datasets rather than the moderate-sample, survey-based setting of the current project.

CatBoost and Gradient Boosting for Small, Categorical Ecological Data

CatBoost's suitability for this project is supported by literature on gradient boosting handling of categorical and small-sample data. The algorithm's original developers demonstrated that CatBoost's permutation-driven target encoding and ordered boosting were designed specifically to address prediction shift and target leakage in conventional gradient boosting, showing robustness on high-cardinality categorical and heterogeneous data (Prokhorenkova et al., 2018). Separate applied work on small-business loan risk assessment found that CatBoost performs well on small datasets because the entire dataset can be used during training without extensive categorical preprocessing (CatBoost synthetic features study, 2021). However, the evidence is not unanimous: a comparison of Random Forest and gradient boosting machines on small, categorical-heavy construction datasets found that bagging-based Random Forest predictions were more stable than boosting-based GBM predictions on a small, mainly categorical dataset, although GBM still achieved strong performance for some

prediction targets (Cha et al., 2021). This suggests CatBoost's suitability, while well justified, should ideally be benchmarked against simpler ensemble alternatives a limitation the current study's scope could acknowledge.

Business Intelligence Dashboards for Environmental Decision Support

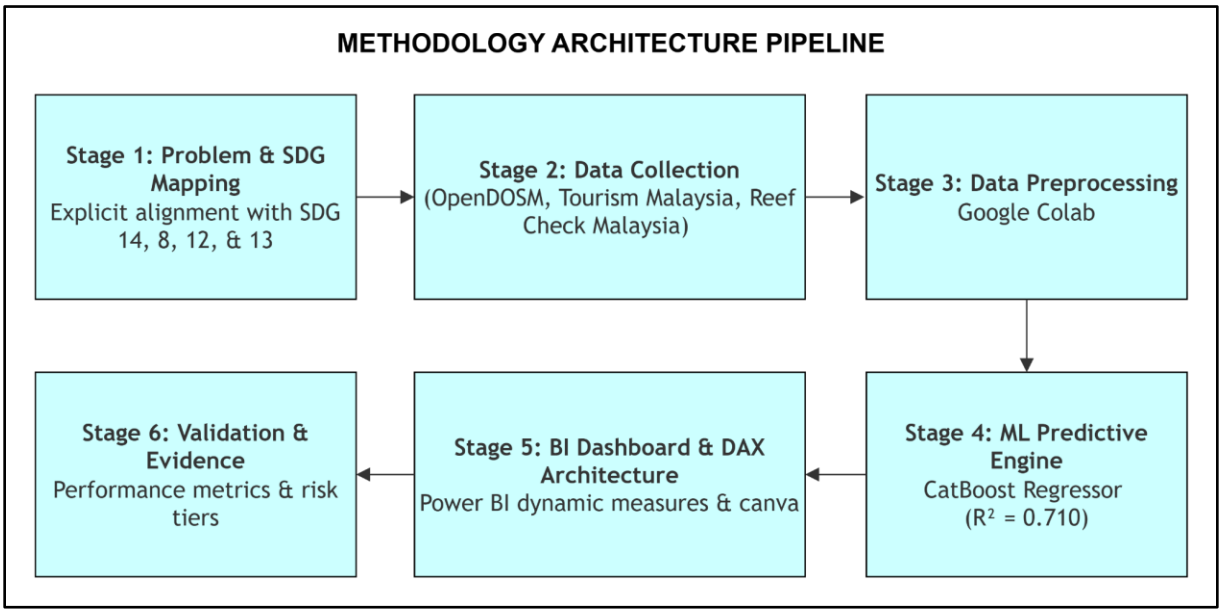
Precedent for linking predictive analytics with executive dashboards exists in the aquaculture and fisheries sector. Raihanto et al. (2023) designed a BI dashboard for real-time water-quality monitoring in a fishery business using a four-stage process scope and plan, analyze and define, architect and design, and build/test/refine and found the resulting system supported the management team's decision-making process, closely paralleling the current project's translation of LCC% forecasts into an executive-facing Power BI interface. Broader literature on BI in environmental management argues that such tools offer valuable guidance for policymakers and researchers seeking to apply business intelligence toward sustainable environmental stewardship (Simon-Ilogho et al., 2024).

Research Gap

Three gaps emerge from this review. First, Malaysian-specific studies consistently document declining reef health but remain retrospective, relying on annual surveys rather than forecasts (The Edge Malaysia, 2025; Mongabay, 2026). Second, ML-based coral forecasting exists internationally but concentrates on remote-sensing or large multi-decadal datasets (Arcodia et al., 2025; PMC, 2024), leaving a gap for survey-based, moderate-sample national contexts such as Malaysia's Reef Check dataset. Third, BI-based decision-support precedents exist in fisheries (Raihanto et al., 2023) but have not been applied to linking coral-reef sustainability forecasts with tourism management authorities in Malaysia specifically. The Blue Lestari project addresses this gap by combining a CatBoost forecasting engine, suited to the small-sample, high-cardinality data documented in the gradient-boosting literature (Prokhorenkova et al., 2018; Cha et al., 2021), with an integrated Power BI decision-support layer modeled on fisheries BI precedent (Raihanto et al., 2023).

METHODOLOGY

To establish an evidence-based, scalable decision-support framework, the **BLUE LESTARI** methodology follows a rigorous 5-stage analytical pipeline. This methodology directly addresses the limitations of traditional, reactive marine conservation by combining machine learning predictive modeling with business intelligence analytics.



Problem Justification & SDG Alignment Mapping

Traditional marine tourism management in Malaysia operates on retrospective field observations. By the time physical reef degradation is confirmed, destination appeal has already deteriorated, directly threatening local coastal economies. The BLUE LESTARI methodology transforms this paradigm from reactive remediation to predictive prevention.

To ensure high societal and environmental impact, the analytical framework is directly mapped to four UN Sustainable Development Goals (SDGs):

- **SDG 14: Life Below Water**
  Forecasts Living Coral Cover (LCC%) trajectories to prevent habitat degradation and establish proactive protection zones.

- **SDG 8: Decent Work & Economic Growth**
  Protects long-term marine tourism yields and coastal employment by preventing ecological collapses at key island hubs.

- **SDG 12: Responsible Consumption & Production**
  Provides dynamic Tourism Carrying Capacity (TCC) analytics to prevent over-tourism during peak seasonal visitor surges.

- **SDG 13: Climate Action**
  Identifies resilient eco-regions and isolates high-risk zones subjected to compounded climate and anthropogenic stress.

Scope of Study

The study scope is structured across specific spatial, temporal, and analytical boundaries to ensure full alignment with competition objectives:

- Spatial Scope: Key marine tourism clusters across Peninsular Malaysia's East Coast (Pulau Redang, Pulau Perhentian, Pulau Tioman, Pulau Tinggi), Malacca Strait (Malacca, Negeri Sembilan, Perak, Kedah) and Borneo (Sabah, Sarawak)
- Temporal Scope: Historical baseline data evaluated up until 2025, with an advanced 2-year predictive forecast horizon spanning 2026–2027.
- Data Sources:
  - Department of Statistics Malaysia (DOSM), *Domestic Tourism Survey by States 2025*.Tourism Malaysia
  - Reef Check Malaysia (RCM), *Annual Coral Reef Survey Reports*.

Data Ingestion & Cleaning

For ingestion, open source data like from OpenDosm or Tourism Malaysia is downloaded directly from the public site. Live Coral Reef Percentage (LCC%) is acquired from Malaysia's Annual Reef Check Survey Reports. It consists of yearly reports starting from 2007 until 2025. However LCC data is limited to only starting 2014. The entire data ingestion, preprocessing, and model development pipeline was built and executed within a Google Colab environment using Python data science libraries.

To build a single source of truth, raw data from various sources and formats were integrated into a single database using common identifiers. Clean and structured tables were created with a proper format, so that all data streams will be connected in an efficient manner.

All text and categorical data were cleansed to address problems associated with different spellings, extra spaces, and different abbreviations. All datasets had standardized naming conventions

and capitalization to ensure consistency in grouping.

Then, raw data points were consolidated into meaningful summaries and metrics. Numerical data points were classified based on relevant geographical and temporal categories to perform calculations such as totals, averages, and ratios, which will uncover any trend.

A quality check was performed to detect data anomalies in terms of duplicates, missing values, and outlier values. All duplicate rows were deleted, missing data were imputed using average values in context, and outlier values were examined using statistical rules to avoid errors in calculations.

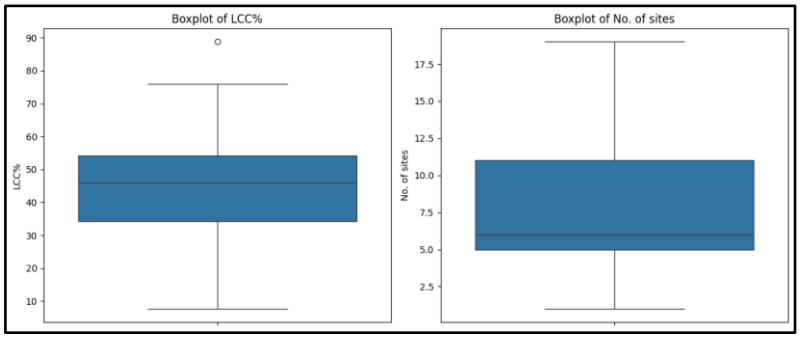
```
data.isnull().sum()
```

	0
Year	0
Eco Region	0
State	0
Areas	0
No. of sites	0
Protection Status	0
LCC%	0

dtype: int64

```
data.duplicated().sum()
```

```
np.int64(0)
```



Finally, proper data types such as numerical, text, and date columns were assigned to each data point in the dataset. The prepared dataset is ready to be used for machine learning algorithms and dashboard visualizations.

	Year	Eco Region	State	Areas	No. of sites	Protection Status	LCC%
0	2014	Sunda Shelf	Terengganu	Perhentian	10	Marine Park	53.62
1	2014	Sunda Shelf	Terengganu	Redang	12	Marine Park	58.13
2	2014	Sunda Shelf	Pahang	Tioman	19	Marine Park	60.60
3	2014	Sunda Shelf	Terengganu	Kapas	4	Marine Park	56.88
4	2014	Sunda Shelf	Terengganu	Bidong and Yu	6	Marine Park	54.58
...	...	...	...	...	...	...	...
376	2025	North Borneo	Sabah	Sipadan	12	Sabah Parks	55.21
377	2025	North Borneo	Sabah	TARP	8	Sabah Parks	13.13
378	2025	North Borneo	Sabah	TMP	18	Sabah Parks	40.35
379	2025	North Borneo	Sabah	TSMP	12	Sabah Parks	51.77
380	2025	North Borneo	Sabah	Usukan Cove	6	No protection	15.94

381 rows × 7 columns

Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) was carried out on the Live Coral Cover (LCC) dataset after cleaning to gain a better understanding of the nature, distribution and hidden patterns of the data before performing any machine learning modeling on it. The EDA process was essential to analyse the variability of LCC% in light of different temporal, geographical, as well as other contextual variables captured in the dataset.

Initially, the general distribution of the LCC% variable was considered by constructing a histogram and calculating the descriptive statistics such as mean, median, standard deviation, minimum, maximum and quartiles. This helped in obtaining an understanding of the central tendency, dispersion and range of the live coral cover data. Next, the temporal pattern was studied by computing the mean of the LCC% data each year from 2014 through 2025 and plotting the data points on a line graph.

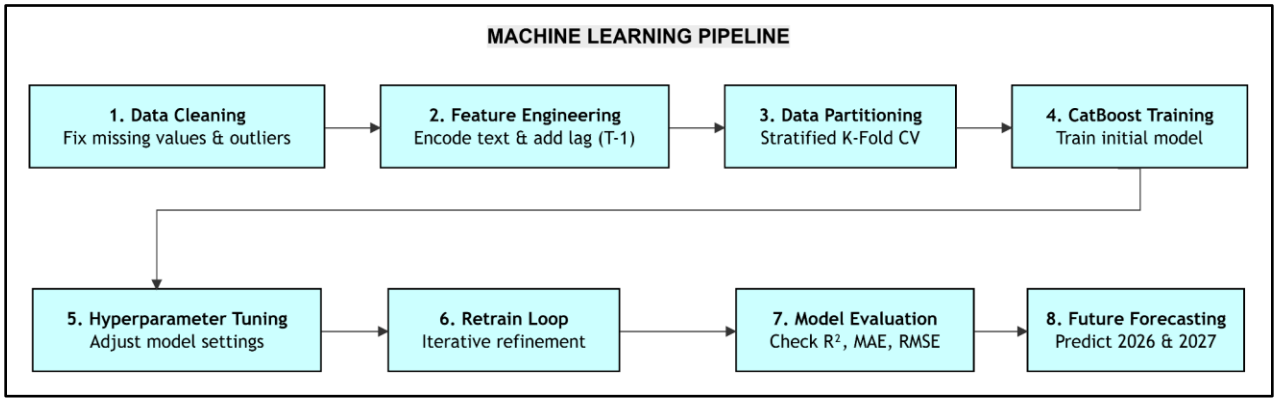
Other comparisons included differences in geographical and contextual factors as well through comparing LCC% for different states, eco regions and protection status. Grouped descriptive statistics and box plots were applied in order to see how the distribution and variation of live coral cover differ in terms of these variables. The sample size in each of the groups was taken into consideration as well as the dataset had unequal representation in some groups.

Finally, an analysis between the number of monitoring stations and LCC% was done via a scatter plot and Pearson correlation. The analysis was carried out to determine the direction and the level of association between the two numeric variables. Generally, EDA gave us insights about the historic LCC dataset and its features before moving to feature engineering and modeling.

Machine Learning Pipeline

In this project, the Machine Learning implementation is done for predicting the coral reefs sustainability for future of Malaysian tourism. Specifically, we use CatBoost Regressor model to predict Live Coral Coverage Percentage (LCC%) for a total of 56 marine tourism areas across Peninsular Malaysia and Borneo. Our main goal is to forecast the coral behaviors in Malaysia for the next 2 years, 2026 and 2027 as well as pinpointing the risky areas.

CatBoost (Categorical Boosting) algorithm is used for the modelling section of this project. In the methodology, the choice of the regressor as the main predictor in preference to the usual linear regressor and the famous Random Forest techniques was done due to the following three advantages. Firstly, target encoding of the categorical variables through CatBoost allows us to avoid overfitting when using categorical spatial variables such as Eco-Region and Areas. Secondly, the ability of modeling non-linear dynamics of ecosystems by using gradient boosting decision tree algorithms is a very advantageous feature since we are dealing with coral behaviours in marine systems. Finally, the model offers enhanced robustness to noise through out-of-fold estimation, which eliminates target leakage during iterative gradient boosting. In short, CatBoost is used because we want to overcome the small number of rows and high cardinality of the LCC data, which might caused to an unreliable model.



The machine learning pipeline begins with comprehensive data cleaning, where raw datasets are prepared by removing duplicates, handling missing values, standardizing column names, and eliminating extreme outliers to ensure high data quality. Following cleaning, feature engineering converts categorical text variables such as island names into numerical formats and incorporates an annual lag feature (T-1) to capture historical yearly trends over time. The refined dataset is then partitioned using Stratified K-Fold Cross-Validation, ensuring every split maintains a balanced

representation of key target variables to eliminate subset bias.

Next, the algorithm processes the training data to learn the core relationships between natural capital metrics and tourism activities, building an initial baseline CatBoost model. To maximize predictive power, the model undergoes hyperparameter tuning, where internal parameters such as learning rate, tree depth, and iteration counts are systematically adjusted to prevent overfitting. The CatBoost model then enters an iterative retraining loop using these optimized parameters until it reaches peak stability.

Finally, the completed model is thoroughly evaluated against unseen validation data using three standard performance metrics: the Coefficient of Determination (R2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Once validated, the CatBoost engine generates environmental and economic predictions for the target horizon years of 2026 and 2027.

The outcomes of this model will then be deployed and visualised in Power BI dashboard later.

Business Intelligence Architecture & Power BI Engineering

The Business Intelligence (BI) tier is designed with the aid of Power BI Desktop to function as an executive decision-making support system where raw data streams, statistics analysis, and machine learning outputs are combined into an interactive 2-page visual dashboard. Power BI is leveraged to turn model predictions into comprehensible findings which can be interpreted by non-technical persons and policy evaluators. The solution works locally, within a standalone environment (.pbix), which guarantees speedy data processing and evaluation capabilities as well as absence of any licensing issues connected to using the cloud platform. Custom Data Analysis Expressions (DAX) were developed within the data model to enable dynamic calculation of KPIs, time aggregations, automated calls for states' risks and customized ranking metrics without changing the underlying data.

The narrative of the dashboard unfolds across two pages in the following order:

Page 1: Overview

Page 2: Tourism Pressure

Page 3: Marine Ecosystem

Page 4: Sustainability Profile

Page 5: ML Prediction

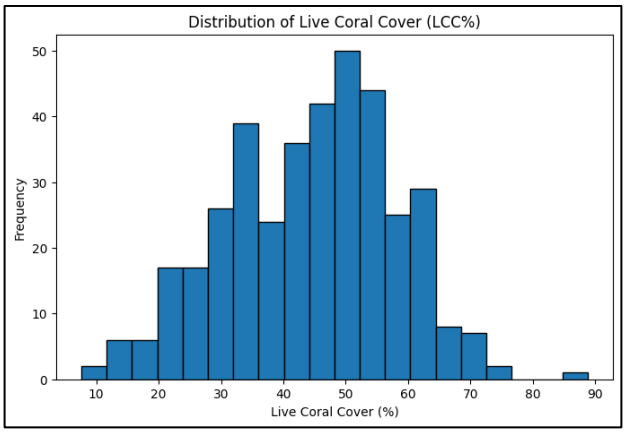
FINDINGS

Exploratory Data Analysis (EDA) Results

Exploratory Data Analysis (EDA) was performed on the cleaned Live Coral Cover (LCC) dataset to investigate the distribution and relationships of the key variables before modeling using machine learning. The exploratory analysis was carried out on LCC%, which is the key variable of interest in this study. Patterns in the data were determined by means of descriptive statistics and visualisations. The correlation between the number of monitoring stations and LCC% was analysed through Pearson correlation analysis.

1. **Distribution of Live Coral Cover**

An analysis of the distribution of LCC% was done in order to provide information about the level and variation in the live coral cover captured in the dataset. This was done by creating a histogram and descriptive statistics of LCC%.




```
data['LCC%'].describe()
```

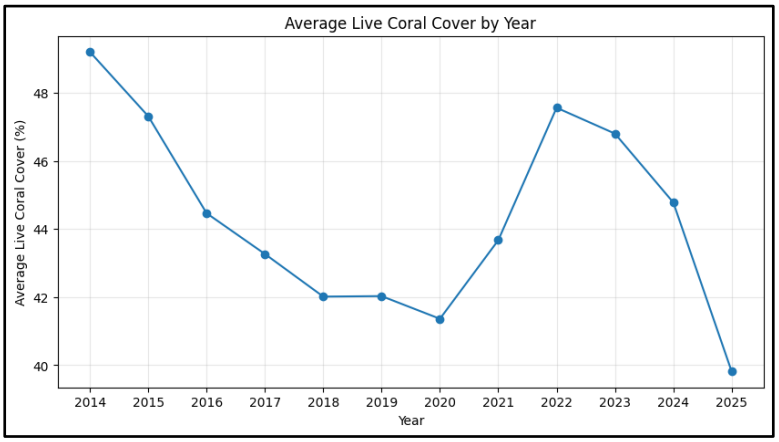
	LCC%
count	381.000000
mean	44.439915
std	13.648436
min	7.630000
25%	34.320000
50%	45.940000
75%	54.170000
max	88.750000

dtype: float64

Based on the figures above, the dataset included a total of 381 observations of the LCC, and the mean and median of LCC for the whole dataset were 44.44% and 45.94%, respectively. The fact that there is only a slight gap between the two suggests that the distribution is not significantly skewed. Most of the observations lie within 30% and 60% of LCC, whereas values vary from 7.63% to 88.75%. The standard deviation of 13.65 percentage points suggests that there is a lot of variability in the live coral coverage among the data. The middle 50% of the data was observed between 34.32% and 54.17%.

2. **Temporal Trend of Live Coral Cover**

Analysis of annual average LCC% from 2014 through 2025 was conducted to see how the live coral cover had been changing during the study period. The average annual values of LCC% were calculated and plotted on a time-series graph.



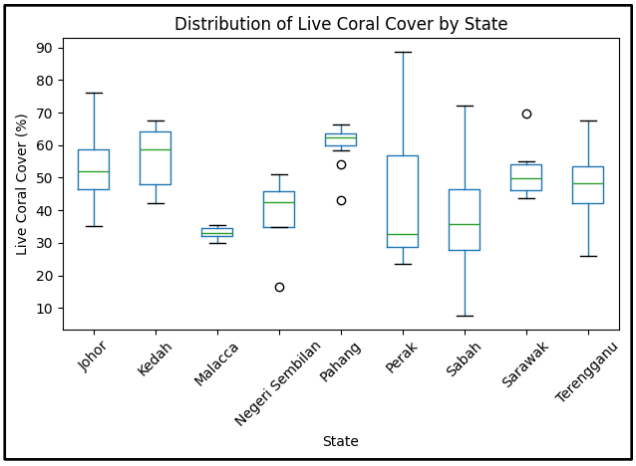
yearly_lcc.round(2)

	Year	LCC%
0	2014	49.20
1	2015	47.31
2	2016	44.46
3	2017	43.27
4	2018	42.01
5	2019	42.03
6	2020	41.36
7	2021	43.67
8	2022	47.56
9	2023	46.80
10	2024	44.78
11	2025	39.82

Based on the visualisations above, it can be seen that LCC% did not demonstrate any linear relationship over the period. The mean LCC decreased from 49.20% in 2014 to 41.36% in 2020. After which it increased to 43.67% in 2021 and further to 47.56% in 2022. Subsequently, the mean LCC% decreased again to 46.80% in 2023, 44.78% in 2024, and 39.82% in 2025, whereby 2025 marked the minimum annual mean in the study period. In general, the findings show that there is year-to-year variability in live coral cover, which justifies the incorporation of temporal data in future models.

3. Live Coral Cover by State

Geographical differences in terms of live coral cover were examined by comparing LCC% in the states where data is available from. To represent the distribution of the states' data, boxplot graph was employed and descriptive statistics were used to compare the states' means, medians, variations and number of observations.




```
state_lcc = (
    data.groupby('State')['LCC%']
      .agg(['count', 'mean', 'median', 'std'])
      .round(2)
      .sort_values('mean', ascending=False)
)

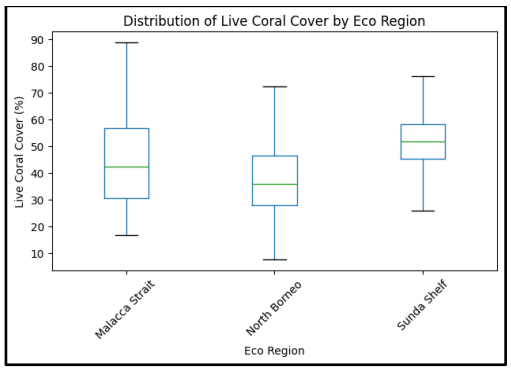
state_lcc
```

State	count	mean	median	std
Pahang	15	60.56	62.50	5.68
Kedah	8	56.23	58.82	9.61
Johor	87	52.69	52.03	9.11
Sarawak	10	51.27	49.76	7.59
Terengganu	69	47.28	48.28	9.85
Perak	20	43.84	32.86	18.80
Negeri Sembilan	5	38.19	42.50	13.41
Sabah	162	36.92	35.94	12.71
Malacca	5	33.13	33.13	2.15

Based on the visualisations above, there were significant variations in live coral cover among the various states featured in the dataset. The highest mean LCC% was recorded for Pahang at 60.56%, followed by Kedah with 56.23%, Johor with 52.69%, Sarawak with 51.27%, Terengganu with 47.28% and Perak with 43.84%. On the other hand, the mean LCC% for Negeri Sembilan, Sabah and Malacca are 38.19%, 36.92% and 33.13% respectively. There was also a difference in the distribution of the data within each individual state. As an illustration, the state of Perak had a standard deviation of 18.80 percentage points, which was a large amount of dispersion compared to only 5.68 percentage points for Pahang. However, the sample size of observation was not uniformly distributed in terms of the state population. Sabah had a total of 162 observations, whereas Negeri Sembilan and Melaka had a mere five observations each. Hence, such an average cannot be regarded as an accurate comparison of coral condition in Malaysia based on the state level.

4. Live Coral Cover by Eco Region

Further analysis of the geography was performed by analysing the LCC% based on eco region. This gives an extended ecological classification of the sampling points beyond the state lines. The distribution and descriptive statistics of the LCC% were analysed among the three eco regions present in the dataset.




```
eco_lcc = (
    data.groupby('Eco Region')['LCC%']
        .agg(['count', 'mean', 'median', 'std'])
        .round(2)
        .sort_values('mean', ascending=False)
)

eco_lcc
```

Eco Region	count	mean	median	std
Sunda Shelf	181	51.20	51.75	9.79
Malacca Strait	38	44.29	42.40	16.48
North Borneo	162	36.92	35.94	12.71

Based on the visualisations above, different values in LCC% were noted among the three eco regions. In terms of LCC%, the Sunda Shelf had the highest mean at 51.20% with a median of 51.75%. Next was the Malacca Strait that had a mean of 44.29% and a median of 42.40%. North Borneo had the lowest mean LCC% at 36.92% and median of 35.94%. Variation was also region-specific. The Malacca Strait region had the highest variation with a standard deviation of 16.48 percentage points, while that of North Borneo and the Sunda Shelf were 12.71 and 9.79 percentage points, respectively. The dataset had a total of 181 Sunda Shelf observations, 162 North Borneo observations, and 38 Malacca Strait observations. This clearly shows that there are geographical variations in LCC observation and it is essential to consider eco region as geographical information for modeling purposes.

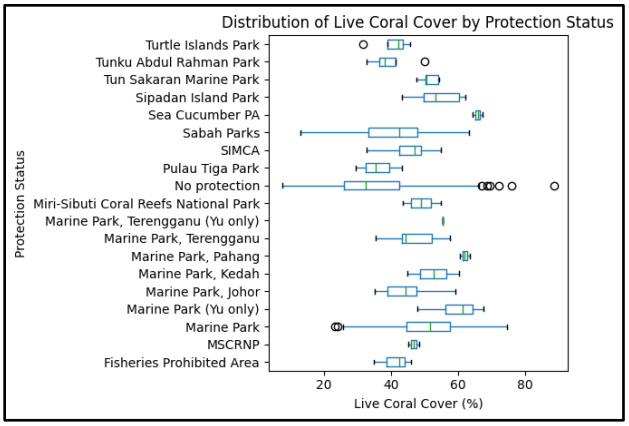
5. **Live Coral Cover by Protection Status**

LCC% was also analyzed based on the protection level of each sample. The analysis was done in order to ascertain if the distribution of live coral cover varied across the protection levels contained in the data set.

```
data['Protection Status'].value_counts()
```

Protection Status	count
Marine Park	156
No protection	119
Sabah Parks	34
SIMCA	12
Marine Park, Johor	10
Marine Park, Terengganu	9
Tun Sakaran Marine Park	5
Sipadan Island Park	5
Turtle Islands Park	4
Miri-Sibuti Coral Reefs National Park	4
Tunku Abdul Rahman Park	4
Marine Park (Yu only)	4
Pulau Tiga Park	3
Fisheries Prohibited Area	3
Marine Park, Pahang	2
MSCRNP	2
Marine Park, Kedah	2
Sea Cucumber PA	2
Marine Park, Terengganu (Yu only)	1

Live coral cover exhibited great variation among protection categories. Nevertheless, the distribution of observations among the protection status categories was extremely unbalanced. The sample included mostly Marine Park (n = 156) and No protection (n = 119). Also, some protection categories had less than five observations.




```
protection_lcc = (
    data.groupby('Protection Status')['LCC%']
        .agg(['count', 'mean', 'median', 'std'])
        .round(2)
        .sort_values('mean', ascending=False)
)

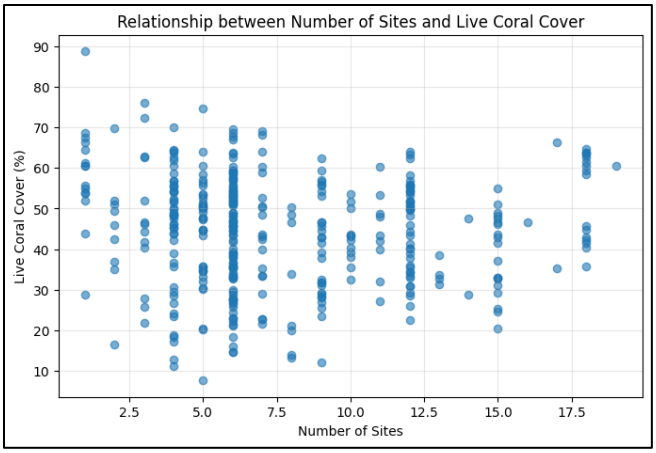
protection_lcc
```

Protection Status	count	mean	median	std
Sea Cucumber PA	2	65.94	65.94	2.21
Marine Park, Pahang	2	62.14	62.14	2.18
Marine Park (Yu only)	4	59.61	61.40	8.44
Marine Park, Terengganu (Yu only)	1	55.52	55.52	NaN
Sipadan Island Park	5	53.91	53.33	7.73
Marine Park, Kedah	2	52.69	52.69	11.05
Tun Sakaran Marine Park	5	51.43	50.68	2.84
Marine Park	156	50.78	51.60	10.12
Miri-Sibuti Coral Reefs National Park	4	49.12	48.96	4.87
MSCRNP	2	46.88	46.88	2.43
Marine Park, Terengganu	9	46.38	44.50	7.63
SIMCA	12	45.48	47.06	6.28
Marine Park, Johor	10	44.26	44.54	7.24
Fisheries Prohibited Area	3	41.15	42.50	5.59
Turtle Islands Park	4	40.44	42.16	6.11
Sabah Parks	34	39.95	42.56	12.69
Tunku Abdul Rahman Park	4	39.80	38.14	7.42
Pulau Tiga Park	3	36.22	35.63	6.95
No protection	119	35.47	32.67	15.01

Of the two largest categories, the average percentage of LCC% for the category Marine Park was 50.78%, while the category No protection had 35.47%. In terms of variation, the standard deviation in the No protection category is 15.01 percentage points, while in the Marine Park category, it is 10.12 percentage points. Some of the other small categories of protection showed higher LCC means, for example, Sea Cucumber PA (65.94%) and Marine Park, Pahang (62.14%). Nevertheless, there were only two records each in these categories. This implies that caution must be exercised when making comparisons using small sample sizes. Overall, these differences merely indicate associations present within the dataset and do not prove causation between the protection category and LCC.

6. **Relationship between Number of Sites and LCC%**

Lastly, the relationship between the number of monitoring sites and LCC% was studied to see if there was any kind of linear relationship between the two numerical variables. The relationship was presented by way of a scatterplot and its strength and direction were measured using the Pearson correlation coefficient.




```
correlation = data['No. of sites'].corr(data['LCC%'])

print(f"Correlation between No. of sites and LCC%: {correlation:.3f}")

Correlation between No. of sites and LCC%: -0.046
```

Based on the visualisations above, it can be seen that the scatter plot demonstrated variability in the percentage of LCC in different numbers of sites. However, there was no observable linear relationship in terms of increase or decrease with respect to number of sites. The Pearson's correlation analysis also yielded an r value of r = -0.046. This indicates that the number of monitoring sites is not linearly related to the LCC%. Nevertheless, it does not mean that the particular variable has no predictive power, since the relationship between the variables under analysis will change depending on the other geographical, temporal and other factors.

In summary, there was considerable time- and geographically-dependent variation in LCC% in the EDA. There were also differences by the protection status categories, although the number of monitoring sites was not linearly correlated with LCC%. This helped gain insight into the dataset before proceeding with feature engineering and building models.

Machine Learning Forecast Model for Coral Reefs Sustainability

The CatBoost model that are built for predicting LCC% evaluated using three main performance metrics which are R-squared, Mean Absolute Error (MAE), Root Mean Square Error (RMSE). The

evaluation are as follows:

Metrics	Performance
R^2	0.710
MAE	5.48%
RMSE	6.78%

The predictive performance metrics demonstrate strong model reliability and robustness in forecasting Living Coral Cover percentage (LCC%). The Coefficient of Determination ($R^2 = 0.710$) indicates that the CatBoost model successfully explains 71% of the variance in LCC% across the studied island ecosystems. In ecological modeling, where biological systems are inherently non-linear and complex, an exceeding 0.70 reflects a high capacity to capture underlying environmental patterns and seasonal trends. The error metrics further validate the practical utility of the model. The Mean Absolute Error (MAE) stands at 5.48%, meaning that predictions deviate from actual field measurements by an average of only 5.48 percentage points. This margin is well within an acceptable threshold for marine science applications, where natural variability in living organisms and oceanographic conditions introduces inherent noise. Additionally, the Root Mean Squared Error of 6.78% confirms that the model prediction errors are consistently distributed with minimal impact from extreme outliers or anomalous errors.

Forecast

This model learns patterns across 12 years of LCC% data in marine areas, helping us to forecast and expect how the graph would look like for the next 2 years. Below is result of the forecast.

```
print(forecast_output_cleaned.head())

   Year    Eco Region          Areas  LCC 2025  Forecasted LCC% Coral_Rating
0  2026   Sunda Shelf  Bidong and Yu      48.0            50.7         Good
1  2026   Sunda Shelf           Gual      45.3            50.0         Good
2  2026   Sunda Shelf       Harimau       49.0            49.9         Fair
3  2026   Sunda Shelf         Hujung      41.8            46.6         Fair
4  2026  North Borneo       Kapalai      14.7            19.8         Poor

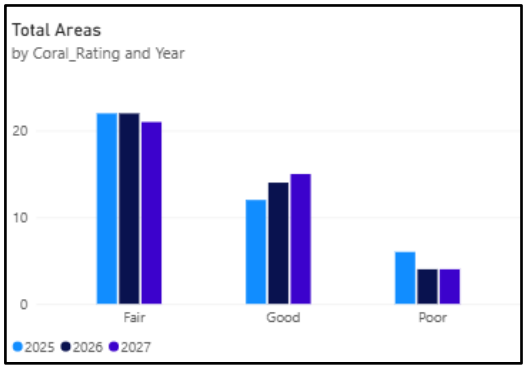
print(forecast_output_cleaned.tail())

    Year    Eco Region          Areas  LCC 2025  Forecasted LCC% Coral_Rating
75  2027  North Borneo           TSMP      45.9            48.6         Fair
76  2027   Sunda Shelf       Tenggol      51.0            48.9         Fair
77  2027   Sunda Shelf         Tinggi      54.1            53.2         Good
78  2027   Sunda Shelf         Tioman      46.4            49.5         Fair
79  2027  North Borneo  Usukan Cove      20.6            23.4         Poor
```



Line chart above shows the forecasting results for 2026 and 2027. Sky blue solid line shows average LCC% across all 56 areas in Malaysia for 2014-2025 based on the data. Dotted dark blue line shows the projected LCC% for the next 2 years. Based on the chart, the average LCC% experiences a steady decline from 2014 until 2020. This is due to so many factors like weather, ocean temperature, and most importantly, human activities. Then starting 2020, graph shows improvements in the coral coverage. This is due to the Covid-19 pandemic that restricts all movement, including tourism activities. The decline in tourism at this period help coral reefs to heal, all the way to 47.56%. However, graph declines to a historical low of 39.82% coverage at 2025, proving that the species is endangered. Our CatBoost model tracks historical data very closely across standard years. The wider residual gap in 2025 reflects the severe, unprecedented impact of the 4th Global Coral Bleaching Event, where ocean heat stress caused a sharp real-world decline to 39.82%. Because our model uses regularization to prevent overfitting, it smooths extreme climate anomalies, which accounts for the higher residual error in 2025 before projecting the recovery trajectory into 2026 and 2027.

The CatBoost model projects a slightly upward trend across the 2 years forecast horizon. LCC% will rebound to approximately 41.4% in 2026 and continue to increase to 42.1% the next year on average. This is good indicator for the marine sustainability to enhance tourism sustainability. But, authorities still need to monitor this case for Marine tourism in coming years.



Bar chart above displays the number of areas by its coral rating levels which are Fair, Good, Poor and Excellent for 2025 and forecasted years 2026 and 2027. There is not a single island that is rated Excellent for all these years which is concerning. If we look deeper, most areas are rated Fair and less Poor rating. CatBoost model has predicted that the number of Good coral reef coverage will increase over the next 2 years. However, number of Poor areas is still the same, which needs acted for. A total of 4 areas will still be at high risk by 2027.

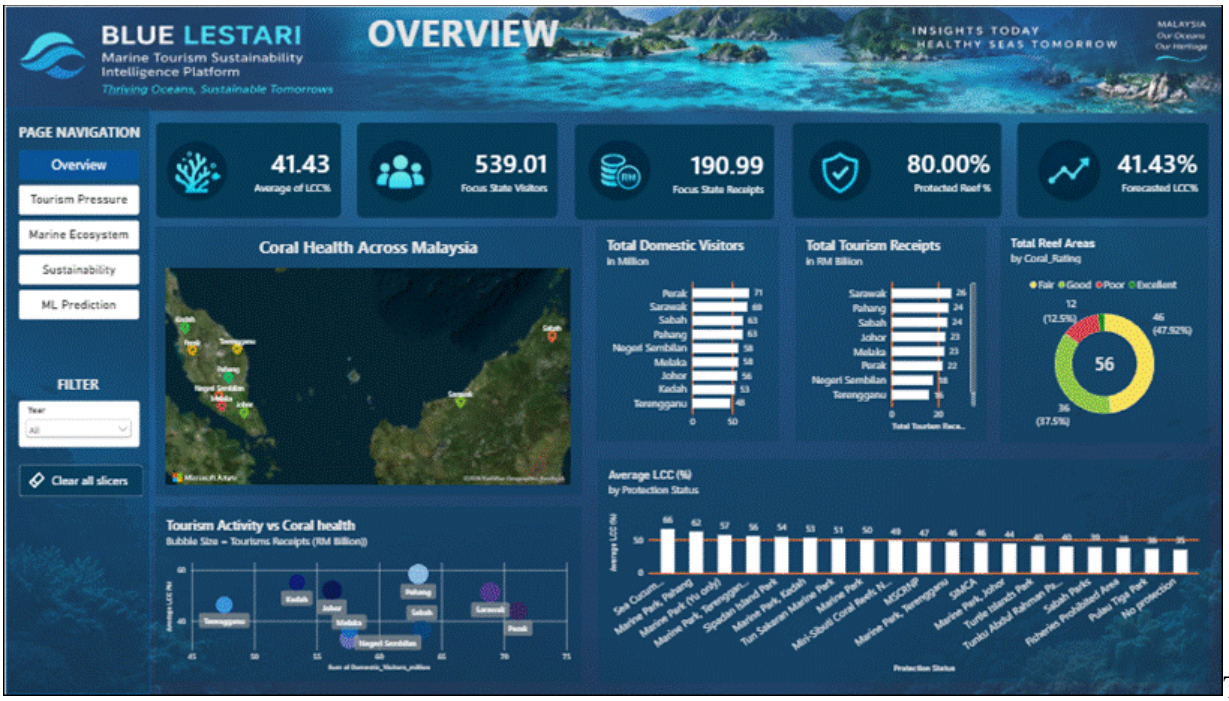
The model highlights the exact state, area, and eco-region that are at high risk, which have the lowest average LCC% among others. The three spatial levels are as follows.

Lowest LCC State	Malacca (32.4%)
Lowest LCC Eco-Region	North Borneo (33.7%)
Lowest LCC Area/Island	Tunku Abdul Rahman Park, TARP (14.1%)

These spatial areas need critical monitoring, authorities could come up with initiative to adjust these numbers. This is a crucial step to sustain marine biodiversity and ensure those areas can still be a good tourist attraction.

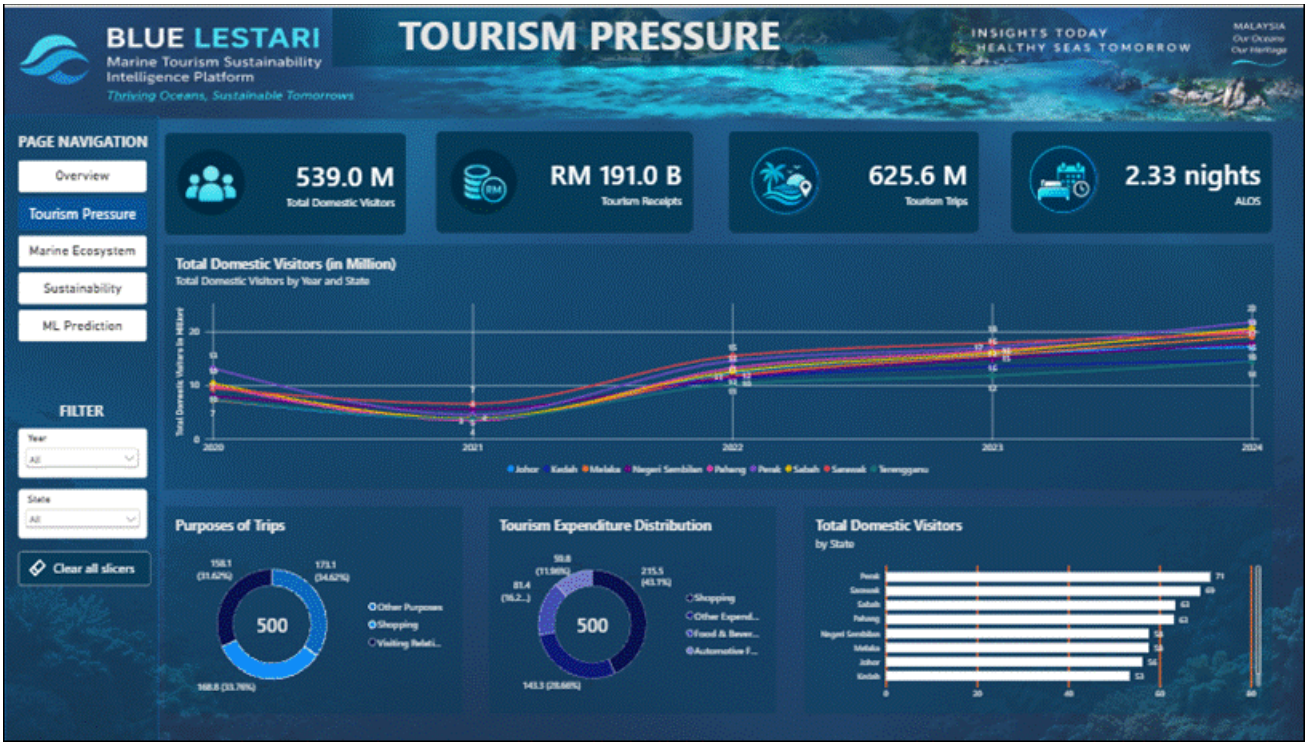
DASHBOARD

Page 1: Overview



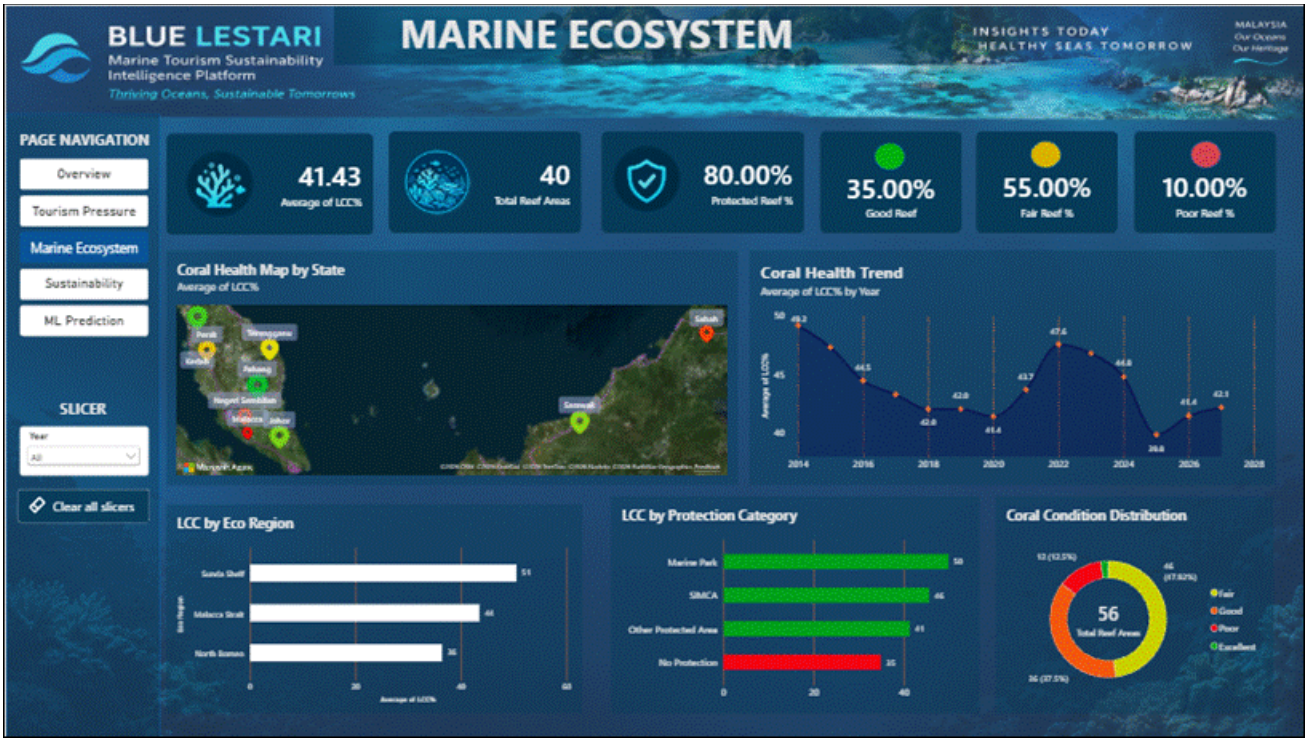
The Overview page highlights differences in tourism activity and coral reef conditions across the selected states. The nine states which are Johor, Kedah, Melaka, Negeri Sembilan, Pahang, Perak, Sabah, Sarawak and Terengganu were selected to provide geographical and ecological representation of Malaysia's key marine tourism areas, covering the East Coast, Malacca Strait and Borneo regions. The selection also considers the availability of comparable tourism and coral reef data at the state level, allowing tourism activity and marine ecosystem conditions to be analysed consistently across locations. This broader coverage enables the study to capture different tourism–reef patterns and provides a more comprehensive basis for supporting marine tourism planning. Perak recorded the highest domestic visitors at approximately 71 million, followed by Sarawak at 69 million and Sabah at 63 million, while Sarawak recorded the highest tourism receipts at approximately RM26 billion. The tourism–coral comparison shows that high tourism activity does not necessarily correspond to lower coral health, as states such as Pahang and Sarawak recorded relatively high tourism activity alongside higher LCC, while Sabah showed lower coral health. The coral health map further highlights spatial differences in reef conditions across the selected states. Overall, the dashboard records an average LCC of 41.43%, with approximately 80% of surveyed reef areas protected. These findings indicate that tourism activity should be assessed alongside coral health and protection status when supporting marine tourism planning. The analysis is limited to the available data period of 2014 to 2025.

Page 2: Tourism Pressure



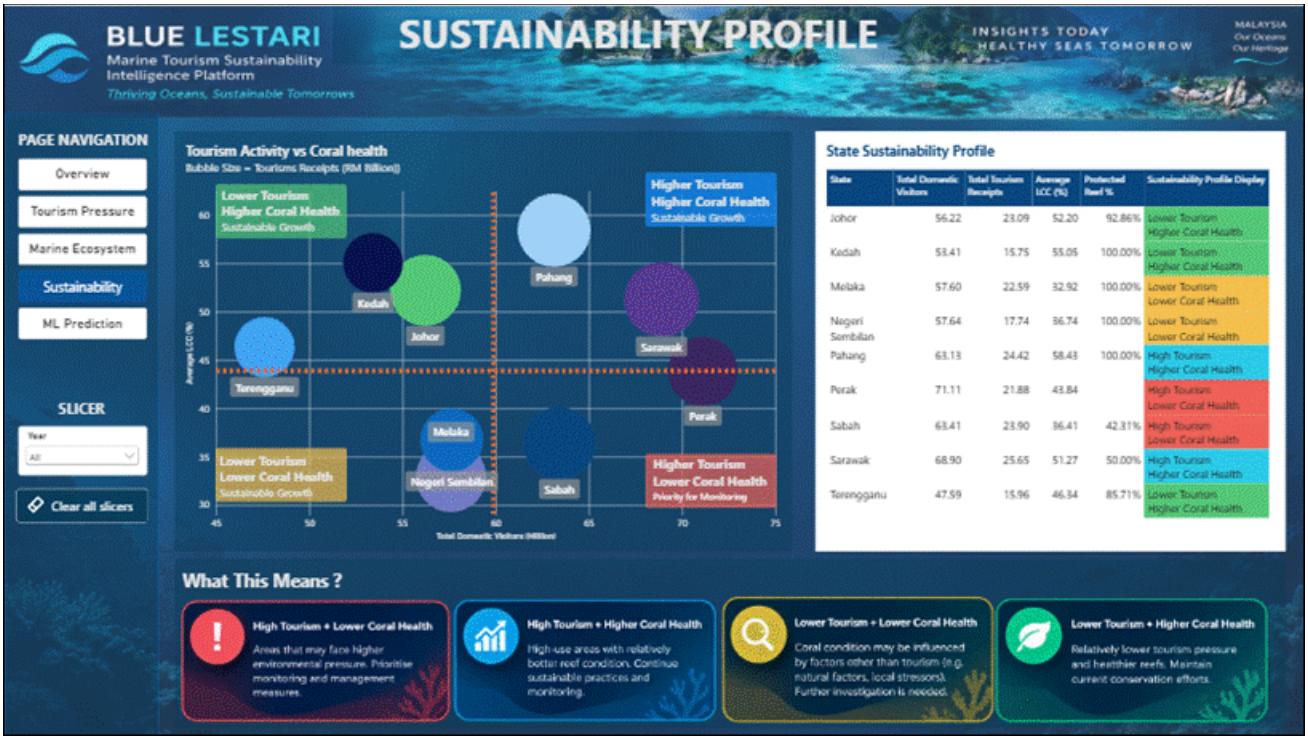
The Tourism Pressure page shows strong domestic tourism activity across the selected states, with 539.0 million visitors, 625.6 million trips, and RM191.0 billion in tourism receipts, while the average length of stay was 2.33 nights. Tourism activity declined in 2021 before recovering from 2022 onwards. In 2024, Perak recorded the highest number of domestic visitors, followed by Sarawak and Sabah. Visiting relatives and friends and shopping were among the main purposes of domestic trips. Overall, the findings highlight the strong economic contribution of tourism while emphasizing the need to manage increasing visitor activity in marine and coastal areas sustainably.

Page 3: Marine Ecosystem



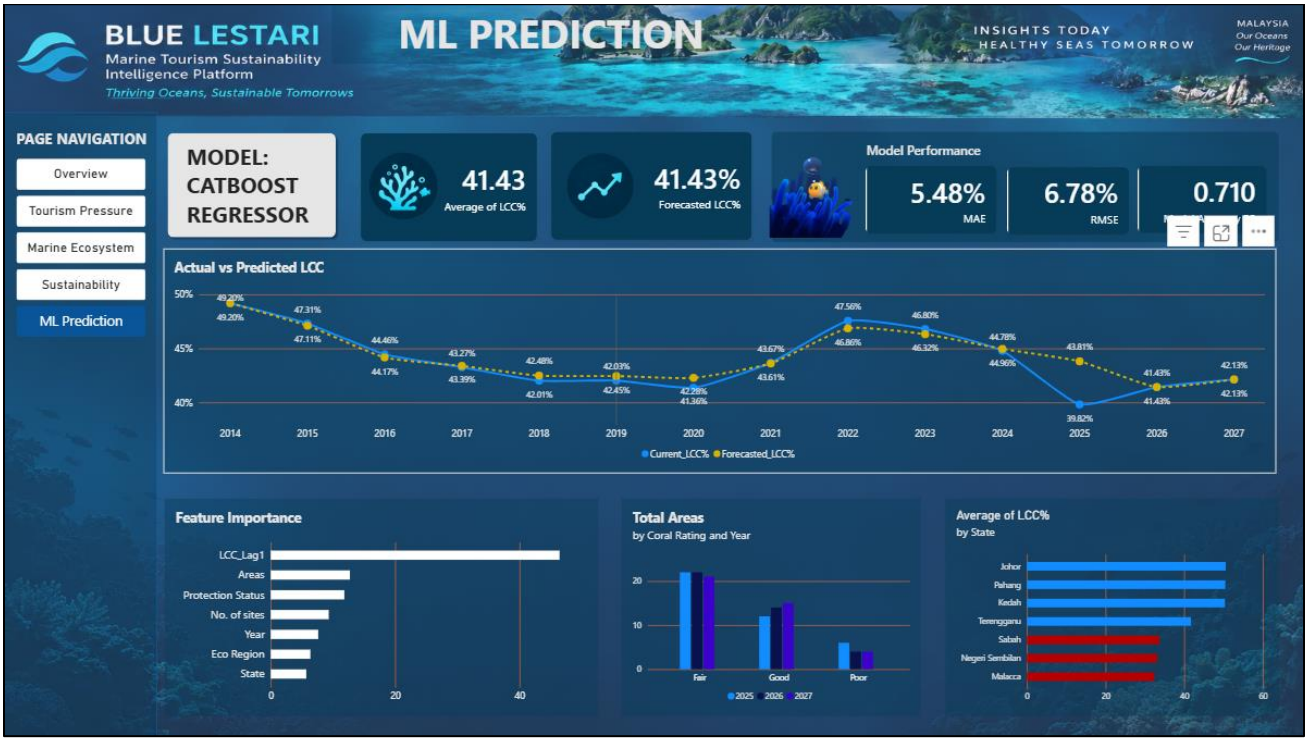
The Marine Ecosystem page shows an overall average LCC of 41.43% across 40 reef areas, with 55% classified as Fair, 35% as Good, and 10% as Poor. Around 80% of reef areas are protected, with Marine Park areas recording a higher average LCC of 50% compared with 35% for areas with no protection. Among the eco-regions, the Sunda Shelf recorded the highest average LCC at 51%, while North Borneo recorded the lowest at 40%. These findings highlight geographical differences in reef health and variations across protection categories. Overall, the results support the need for continued monitoring and targeted conservation efforts, particularly in areas with lower LCC and limited protection. The analysis is based on marine ecosystem data available up to 2025.

Page 4: Sustainability Profile



The Sustainability page integrates tourism activity and coral reef health to develop a state-level sustainability profile. The classification is based on the average domestic visitors and average LCC across the nine selected states, with 59.89 million visitors and 43.98% LCC used as reference values. States above the tourism average are classified as having higher tourism activity, while those below are classified as having lower tourism activity. Similarly, states with LCC above 43.98% are classified as having higher coral health, while those below are classified as having lower coral health. This creates four profiles: High Tourism + Lower Coral Health, High Tourism + Higher Coral Health, Lower Tourism + Lower Coral Health, and Lower Tourism + Higher Coral Health. The profiles provide a descriptive comparison rather than indicating causation, and can help authorities identify areas for closer monitoring, further investigation, and targeted conservation planning.

Page 5: ML Prediction



The ML Prediction page uses a CatBoost Regressor to forecast future Live Coral Cover (LCC), achieving an MAE of 5.48%, RMSE of 0.07, and R^2 of 0.710. LCC_Lag1 was the most influential feature, followed by Areas, Protection Status and survey sites, highlighting the importance of previous coral conditions in predicting future LCC. The predicted trend generally follows the historical LCC pattern, although some differences occur between actual and forecasted values. These predictions provide an additional reference for identifying reef areas that may require closer monitoring but should be interpreted as model-based estimates rather than exact future outcomes.

CONCLUSION

BLUE LESTARI demonstrates how Business Intelligence and Machine Learning can be integrated to support more informed and sustainable marine tourism management in Malaysia. The analysis shows substantial tourism activity across the selected states, alongside considerable geographical and temporal variation in coral reef conditions. The findings also indicate that tourism activity alone does not necessarily correspond to poorer coral health, highlighting the importance of considering tourism pressure together with environmental conditions, protection status and ecological characteristics.

The CatBoost model provides an additional predictive perspective by forecasting future Live Coral Cover (LCC) for 2026 and 2027. With an R^2 of 0.710, MAE of 5.48% and RMSE of 6.78%, the model demonstrates its ability to capture patterns within the available reef data, while the predictions can support the identification of areas that may require closer monitoring.

Through the integrated Power BI dashboard, tourism indicators, coral reef conditions and ML predictions are presented in a single platform, allowing users to explore tourism pressure, marine ecosystem conditions, sustainability profiles and future reef trends. Overall, BLUE LESTARI provides a data-driven foundation for connecting tourism development with marine ecosystem monitoring and supporting more balanced decision-making. However, the findings should be interpreted within the limitations of the available data, which covers the period up to 2025, while the ML forecasts extend to 2026 and 2027.

REFERENCES

Ahmad, S. A., & Hanley, N. (2009). Willingness to pay for reducing crowding effect damages in Marine Parks in Malaysia. *The Singapore Economic Review*, *54*(1), 21–39.

Arcodia, M. C., Karp, R., & Barnes, E. A. (2025). An explainable machine learning prediction system for early warning of heat stress on Florida's Coral Reef. *Environmental Research Communications*, *7*(12), Article 125019.

Cha, G.-W., Moon, H.-J., & Kim, Y.-C. (2021). Comparison of random forest and gradient boosting machine models for predicting demolition waste based on small datasets and categorical variables. *International Journal of Environmental Research and Public Health*, *18*(16), 8530.

CatBoost model with synthetic features in application to loan risk assessment of small businesses. (2021). *arXiv preprint*. https://arxiv.org/abs/2106.07954

Lim, L. C. (1998). *Carrying capacity assessment of Pulau Payar Marine Park, Malaysia* (BOBP/REP/79). Bay of Bengal Programme, Food and Agriculture Organization of the United Nations.

Machine-learning models of coral cover and life histories reveal that climate refugia for coral reefs persist into 2050. (2026). *ResearchGate*. Retrieved September 2026.

Mongabay. (2026, February 5). *Malaysia lost 20% of its coral reefs in three years*. Mongabay News. https://news.mongabay.com/short-article/2026/02/malaysia-lost-20-of-its-coral-reefs-in-three-years/

A generalized machine learning model for long-term coral reef monitoring in the Red Sea. (2024). *PMC / ScienceDirect*.

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*.

Raihanto, M. S., Febrianti, M. A., Qurtubi, Setiawan, D., & Auliana, W. (2023). Designing business intelligence dashboards to support decision-making in a fishery business. *Buletin Ilmiah Sarjana Teknik Elektro (BISTE)*, *5*(4). https://doi.org/10.12928/biste.v5i4.9207

Reef Check Malaysia. (2025). *Reefs and tourism: Balancing economic benefits against community & ecosystem impacts*. https://reefcheck.org.my/reefs-and-tourism-balancing-economic-benefits-against-community-amp-ecosystem-impacts/

Simon-Ilogho, B., Kehinde, K., & Kehinde, T. (2024). The role of business intelligence in enhancing environmental management strategies. *Arabian Journal of Business and Management Review (Kuwait Chapter)*, *13*(2), 25–31.

The Edge Malaysia. (2025, March 17–23). Marine: Declining reef health in Malaysia requires action. *The Edge Malaysia Weekly*. https://theedgemalaysia.com/node/747879

Zhang, L. Y., Chung, S. S., & Qiu, J. W. (2016). Ecological carrying capacity assessment of diving site: A case study of Mabul Island, Malaysia. *Journal of Environmental Management*, *183*, 253–259.